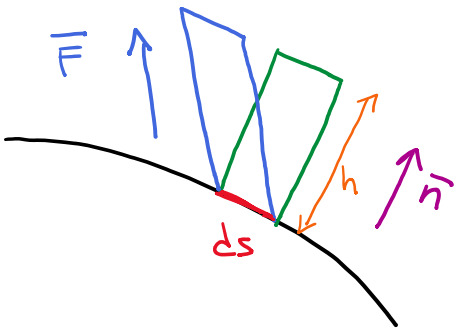
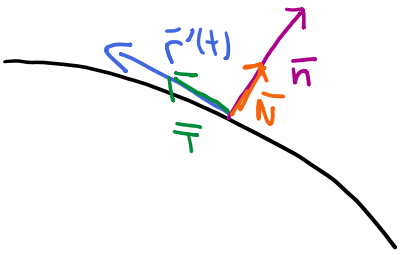
Problema 1

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Demuestra la fórmula para calcular el flujo de un fluido a través de un arco de curva.





$\bar{r}(t) = (x(t), y(t)) \longrightarrow \bar{r}'(t) = (x'(t), y'(t)) \longrightarrow \|\bar{r}'(t)\| = \sqrt{(x'(t))^2 + (y'(t))^2}$

$$\bar{T}(t) = \frac{\bar{r}'(t)}{\|\bar{r}'(t)\|} = \frac{(x'(t), y'(t))}{\sqrt{(x'(t))^2 + (y'(t))^2}}$$

Giro de 90° sobre $\bar{r}'(t)$: $\bar{n}(t) = (-y'(t), x'(t)) \longrightarrow \|\bar{n}(t)\| = \sqrt{(x'(t))^2 + (y'(t))^2}$

$$\text{Flujo} = h \cdot ds = \|\text{proj}_{\bar{n}} \bar{F}\| \cdot ds = |\bar{F} \cdot \bar{N}| \cdot ds = \frac{|\bar{F} \cdot \bar{n}|}{\|\bar{n}\|} \cdot ds = \frac{|\bar{F} \cdot \bar{n}|}{\|\bar{n}\|} \cdot \|\bar{r}'(t)\| \cdot dt =$$

$$= \frac{|\bar{F} \cdot \bar{n}|}{\|\bar{r}'(t)\|} \cdot \|\bar{r}'(t)\| \cdot dt = |\bar{F} \cdot \bar{n}| \cdot dt$$

Por ello, $$\text{Flujo} = \int_C \bar{F} \cdot \bar{N} \cdot ds = \int_{t_a}^{t_b} \bar{F}(\bar{r}(t)) \cdot \bar{n}(x(t), y(t)) \cdot dt$$

Problema 2

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Determina el valor de la integral $\int_C \overline{F} \cdot d\overline{r}$ con $\overline{F}(x,y,z) = \left(\frac{z}{x^2y} - \frac{z}{x^2+z^2}, \frac{z}{xy^2}, \frac{x}{x^2+z^2} - \frac{1}{xy}\right)$ para cualquier arco de curva C que comience en el punto $P = \left(1, \frac{1}{3}, 1\right)$ y termine en $Q = \left(2, \frac{1}{2}, 2\right)$, estando toda ella contenida en el primer octante.

Vamos a comprobar que el campo $\overline{F}(x,y,z) = \left(\underbrace{\frac{z}{x^2y} - \frac{z}{x^2+z^2}}_{P(x,y,z)}, \underbrace{\frac{z}{xy^2}}_{Q(x,y,z)}, \underbrace{\frac{x}{x^2+z^2} - \frac{1}{xy}}_{R(x,y,z)}\right)$ es irrotacional:

$$\frac{\partial P(x,y,z)}{\partial y} = \frac{-x^2z}{(x^2y)^2} = \frac{-z}{x^2y^2} = \frac{-zy^2}{(xy^2)^2} = \frac{\partial Q(x,y,z)}{\partial x}$$

$$\frac{\partial P(x,y,z)}{\partial z} = \frac{1}{x^2y} - \frac{(x^2+z^2-2z^2)}{(x^2+z^2)^2} = \frac{1}{x^2y} - \frac{(x^2-z^2)}{(x^2+z^2)^2} = \frac{z^2-x^2}{(x^2+z^2)^2} + \frac{y}{(xy)^2} = \frac{\partial R(x,y,z)}{\partial x}$$

$$\frac{\partial Q(x,y,z)}{\partial z} = \frac{1}{xy^2} = \frac{x}{(xy)^2} = \frac{\partial R(x,y,z)}{\partial y}$$

Como podemos utilizar una curva que une los puntos $(1,\frac{1}{3},1)$ y $(2,\frac{1}{2},2)$ en un conjunto convexo del primer octante, y las funciones P, Q, R son de clase 1 en dicho conjunto convexo, el campo es conservativo en ese conjunto y podemos utilizar la función potencial.

- $\frac{\partial U(x,y,z)}{\partial y} = Q(x,y,z) \Rightarrow U(x,y,z) = \int \frac{z}{xy^2}\,dy + C(x,z) = \frac{-z}{xy} + C(x,z)$

- $\frac{\partial U(x,y,z)}{\partial x} = P(x,y,z) \Rightarrow \frac{yz}{x^2y^2} + \frac{\partial C(x,z)}{\partial x} = \frac{z}{x^2y} - \frac{z}{x^2+z^2} \Rightarrow$

$$\Rightarrow C(x,z) = -\int \frac{z}{x^2+z^2}\,dx + C(z) = -\int \frac{z}{z^2\left(1+\frac{x^2}{z^2}\right)}\,dx + C(z) =$$

$$= -\int \frac{1/z}{1+\left(\frac{x}{z}\right)^2}\,dx + C(z) = -\operatorname{arctg}\left(\frac{x}{z}\right) + C(z) \Rightarrow U(x,y,z) = -\frac{z}{xy} - \operatorname{arctg}\left(\frac{x}{z}\right) + C(z)$$

- $\dfrac{\partial U(x,y,z)}{\partial z} = R(x,y,z) \Rightarrow -\dfrac{1}{xy} - \left(\dfrac{-\frac{x}{z^2}}{1+\frac{x^2}{z^2}}\right) + C'(z) = \dfrac{x}{x^2+z^2} - \dfrac{1}{xy} \Rightarrow$

$$\Rightarrow C'(z) = 0 \Rightarrow C(z) = C \Rightarrow \boxed{U(x,y,z) = -\frac{z}{xy} - \operatorname{arctg}\left(\frac{x}{z}\right) + C}$$

Por lo tanto:

$$\int_C \bar{F}\cdot d\bar{r} = U(B) - U(A) = U\left(2, \tfrac{1}{2}, 2\right) - U\left(1, \tfrac{1}{3}, 1\right) =$$

$$= \left(-\frac{2}{2\cdot\frac{1}{2}} - \operatorname{arctg}\left(\frac{2}{2}\right)\right) - \left(-\frac{1}{1\cdot\frac{1}{3}} - \operatorname{arctg}\left(\frac{1}{1}\right)\right) = -2 + 3 = \boxed{1}$$

Alternativamente:

$$\bar{r}(t) = \left(t, \frac{1}{4-t}, t\right),\ t_a = 1,\ t_b = 2,\ \bar{r}'(t) = \left(1, \frac{1}{(4-t)^2}, 1\right)$$

$$\int_C \bar{F}\cdot d\bar{r} = \int_1^2 \left(\frac{t}{t^2\left(\frac{1}{4-t}\right)} - \frac{t}{t^2+t^2}, \frac{t}{t\left(\frac{1}{4-t}\right)^2}, \frac{t}{t^2+t^2} - \frac{1}{t\left(\frac{1}{4-t}\right)}\right)\cdot\left(1, \frac{1}{(4-t)^2}, 1\right)dt =$$

$$= \int_1^2 \left(\frac{1}{t\left(\frac{1}{4-t}\right)} - \frac{1}{2t} + (4-t)^2\cdot\frac{1}{(4-t)^2} + \frac{1}{2t} - \frac{1}{t\left(\frac{1}{4-t}\right)}\right)dt =$$

$$= \int_1^2 dt = [t]_1^2 = \boxed{1}$$

Problema 3

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Calcula el área de la superficie dada por la expresión $z^2 = 2xy$ que se halla comprendida entre los planos $x = 1$, $x = 4$, $y = 9$, $y = 16$ exclusivamente en el primer octante.

Al tratarse del primer octante, $z = \sqrt{2xy}$

$$dS = \sqrt{1 + (f_x(x,y))^2 + (f_y(x,y))^2} = \sqrt{1 + \left(\frac{y}{\sqrt{2xy}}\right)^2 + \left(\frac{x}{\sqrt{2xy}}\right)^2} =$$

$$= \sqrt{1 + \frac{y^2}{2xy} + \frac{x^2}{2xy}} = \sqrt{\frac{x^2+y^2+2xy}{2xy}} = \sqrt{\frac{(x+y)^2}{2xy}} \overset{\text{1er octante}}{=} \frac{x+y}{\sqrt{2xy}}$$

$$\text{Área} = \iint_S dS = \iint_R \frac{x+y}{\sqrt{2xy}}\,dx\,dy = \iint_R \frac{x+y}{\sqrt{2}\,x^{1/2}y^{1/2}}\,dx\,dy =$$

$$= \frac{1}{\sqrt{2}} \iint_R \left(x \cdot x^{-1/2} \cdot y^{-1/2} + x^{-1/2} \cdot y \cdot y^{-1/2}\right) dx\,dy = \frac{1}{\sqrt{2}} \iint_R \left(x^{1/2}y^{-1/2} + x^{-1/2}y^{1/2}\right) dx\,dy =$$

$$= \frac{1}{\sqrt{2}} \int_{y=9}^{y=16} \left(\int_{x=1}^{x=4} \left(x^{1/2}y^{-1/2} + x^{-1/2}y^{1/2}\right) dx \right) dy =$$

$$= \frac{1}{\sqrt{2}} \int_{y=9}^{y=16} \left(y^{-1/2} \left[\frac{x^{3/2}}{3/2}\right]_1^4 + y^{1/2} \left[\frac{x^{1/2}}{1/2}\right]_1^4 \right) dy =$$

$$= \frac{1}{\sqrt{2}} \int_{y=9}^{y=16} \left(\frac{2}{3}\, y^{-1/2} \cdot (8-1) + 2 \cdot y^{1/2} (2-1) \right) dy =$$

$$= \frac{2}{\sqrt{2}} \int_{y=9}^{y=16} \left(\frac{7}{3}\, y^{-1/2} + y^{1/2} \right) dy = \frac{2}{\sqrt{2}} \left(\frac{7}{3} \left[\frac{y^{1/2}}{1/2}\right]_9^{16} + \left[\frac{y^{3/2}}{3/2}\right]_9^{16} \right) =$$

$$= \frac{2}{\sqrt{2}} \left(\frac{14}{3}(4-3) + \frac{2}{3}(64-27) \right) = \frac{2}{\sqrt{2}} \left(\frac{14}{3} + \frac{74}{3} \right) = \frac{176}{3\sqrt{2}} = \boxed{\frac{88\sqrt{2}}{3}}$$

Problema 4

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Dado el campo $\overline{F}(x,y,z) = \left(x^2+y, z, xyz\right)$ calcula $\iint_S \text{rot}(\overline{F})\cdot\overline{N}dS$, donde $S = S_1 \cup S_2$, siendo $S_1 = \left\{(x,y,z)\in\mathbb{R}^3 \mid x^2+y^2+z^2 = 9,\ z\geqslant 0\right\}$ y $S_2 = \left\{(x,y,z)\in\mathbb{R}^3 \mid x^2+y^2 = 9,\ -10\leqslant z\leqslant 0\right\}$.

Si consideramos la superficie $S_3 = \{(x,y,z)\in\mathbb{R}^3 \mid x^2+y^2\leq 9,\ z=-10\}$, entonces la superficie $S^* = S_1\cup S_2\cup S_3$ es cerrada y podemos aplicar el teorema de Gauss:

$$\iiint_D (\nabla\cdot\overline{G})\,dV = \iint_{S^*}\overline{G}(x,y,z)\cdot\overline{N}\,dS = \iint_S \overline{G}(x,y,z)\cdot\overline{N}\,dS + \iint_{S_3}\overline{G}(x,y,z)\cdot\overline{N}\,dS$$

donde $\overline{G} = \nabla\times\overline{F} = \begin{vmatrix} \overline{i} & \overline{j} & \overline{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ (x^2+y) & z & xyz \end{vmatrix} = (xz-1, -yz, -1)$

$$\iiint_D (\nabla\cdot\overline{G})\,dV = \iiint_D (\cancel{z} - \cancel{z} + 0)\,dV = 0$$

$$\iint_{S_3}\overline{G}(x,y,z)\cdot\overline{N}\,dS = \iint_{S_3}(xz-1, -yz, -1)\cdot(0,0,-1)\,dS =$$

$$= \iint_{S_3} dS = \text{Área}(S_3) = \pi\cdot 3^2 = 9\pi$$

$$\iint_{S_1\cup S_2}\text{rot}(\overline{F})\cdot\overline{N}\,dS + \iint_{S_3}\text{rot}(\overline{F})\cdot\overline{N}\,dS = 0 \Rightarrow \iint_{S_1\cup S_2}\text{rot}(\overline{F})\overline{N}\,dS = -\iint_{S_3}\text{rot}(\overline{F})\overline{N}\,dS \Rightarrow$$

$$\Rightarrow \boxed{\iint_{S_1\cup S_2}\text{rot}(\overline{F})\cdot\overline{N}\,dS = -9\pi}$$

Nota: Los vectores normales de S_1 y S_2 están tomados hacia el exterior del volumen que encierra $S_1\cup S_2\cup S_3$.

Alternativamente:

$$\iint_S \text{rot}(\bar{F})\cdot\bar{N}\,dS = \oint_C \bar{F}(x,y,z)\,d\bar{r} = \left| \begin{array}{l} \bar{r}(t) = (3\cos(t), 3\,\text{sen}(t), -10) \\ \bar{r}'(t) = (-3\,\text{sen}(t), 3\cos(t), 0) \\ t_a = 0,\ t_b = 2\pi \end{array} \right| =$$

$$= \int_0^{2\pi} \left(3^2\cos^2(t) + 3\,\text{sen}(t), -10, 3\cos(t)\cdot 3\,\text{sen}(t)(-10)\right)\cdot(-3\,\text{sen}(t), 3\cos(t), 0)\,dt =$$

$$= \int_0^{2\pi} \left(-27\,\text{sen}(t)\cos^2(t) - 9\,\text{sen}^2(t) - 30\cos(t) + 0\right) dt =$$

$$= \int_0^{2\pi} \left(-27\,\text{sen}(t)\cos^2(t) - 9\left(\frac{1-\cos(2t)}{2}\right) - 30\cos(t)\right) dt =$$

$$= \left[27\,\frac{\cos^3(t)}{3} - \frac{9}{2}t + \frac{9}{2}\,\frac{\text{sen}(2t)}{2} - 30\,\text{sen}(t)\right]_0^{2\pi} =$$

$$= -\frac{9}{2}\cdot 2\pi = \boxed{-9\pi}$$